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ECEN 5224

Week 4 Lab Report

02/04/26

Lab 1

Objective:

The objective of this lab is to simulate the reflections of various impedance cases of an ideal 50 Ω transmission line.

Method:

ADS was used to simulate an ideal transmission line in the following circuit. A step voltage source was utilized with a transient simulation. A source resistor was altered between 5 Ω, 50 Ω, and 950 Ω while the load resistance was set to either 50 Ω or 1 MΩ to represent a matched and open circuit. Transient simulations were performed for six cases of the source and load impedances. Voltages at both, the source and receiver end, were graphed.

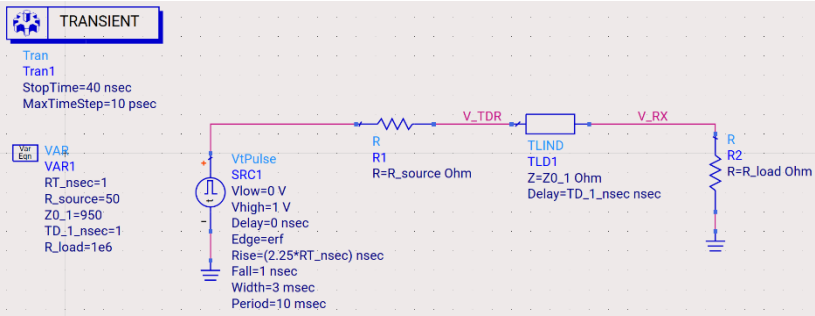


Figure 1: Lab 1 Completed Circuit

Results and Plots:

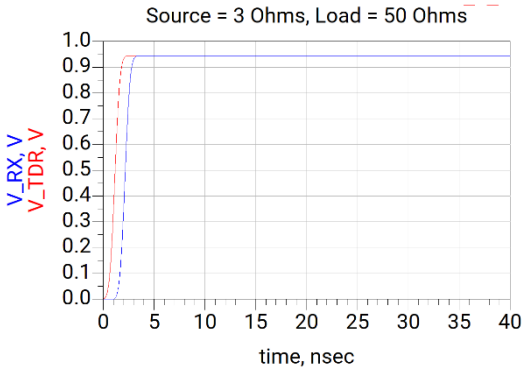


Figure 2: Source Impedance 3 Ω RX Impedance 50 Ω

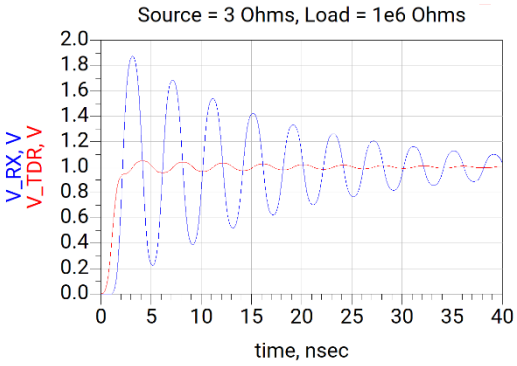


Figure 3: Source Impedance 3 Ω RX Impedance 1MΩ

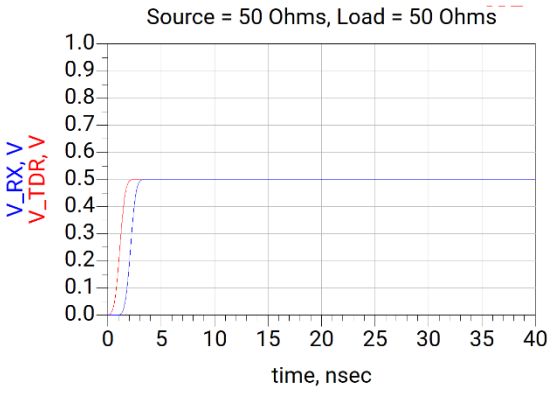


Figure 4: Source Impedance 50 Ω RX Impedance 50 Ω

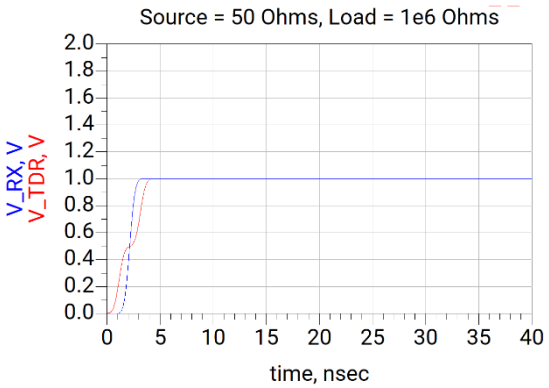


Figure 5: Source Impedance 50 Ω RX Impedance 1MΩ

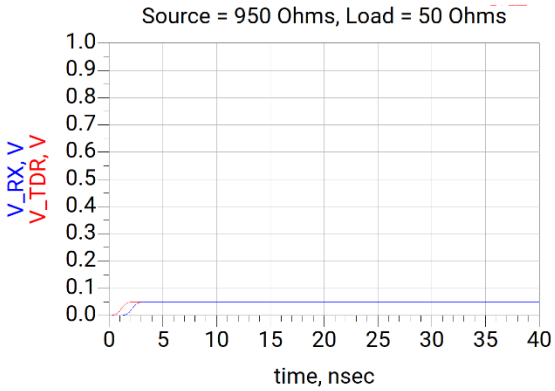


Figure 6: Source Impedance 950 Ω RX Impedance 50 Ω

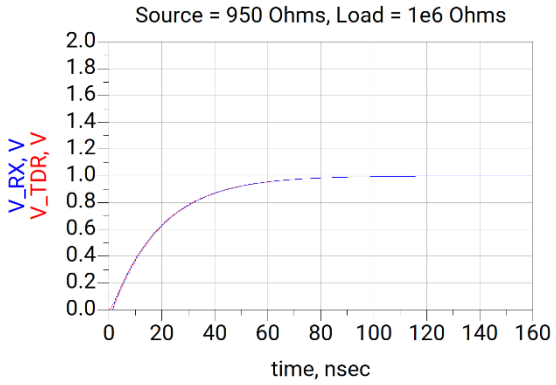


Figure 7: Source Impedance 950 Ω RX Impedance 1MΩ

Analysis and Conclusion:

The simulated plots show that matching impedance heavily affects reflections on a transmission line. When both, the source and load impedances are matched, there are minimal reflections. It provides the best signal as both voltages flattened quickly as seen in Figures 4 and 5. When they are not matched, there is significant ringing and many reflections between both load and source. It can also be seen that it takes longer for the voltages to increase and settle (see Figures 3, 6, and 7).

These simulations prove that matching impedance is critical for stable performance in signal integrity systems. Mismatched systems exhibit ringing and non-ideal voltage behavior which will heavily impact performance illustrating the need for carefully adjusted component values.